

Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Scenario:

A large organization is migrating its on-premises infrastructure to the AWS cloud. The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

Objectives:

Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.

Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

Note:

Create 4 VPCs in 4 different regions and set up Transit Gateway.

Do not use default VPCs.

Prepare a document with all the screenshots and push it to a GitHub repo, then share the URL.

□ Step 1: Create 4 VPCs (No Default VPCs)

We need 4 VPCs in different regions:

- **us-east-1 (Hub)**
- **us-west-2 (Spoke 1)**
- **eu-west-1 (Spoke 2)**
- **ap-south-1 (Spoke 3)**

□ In each region, do the following:

1. Go to **VPC Console** → **Your VPCs** → **Create VPC**.
 - Name: **vpc-hub-us-east-1** (adjust per region).

- IPv4 CIDR: **10.0.0.0/16** (use different CIDRs for each region). Example:
 - Hub (us-east-1): **10.0.0.0/16**
 - Spoke1 (us-west-2): **10.1.0.0/16**
 - Spoke2 (eu-west-1): **10.2.0.0/16**
 - Spoke3 (ap-south-1): **10.3.0.0/16**
 - Tenancy: Default.
2. Subnets: Create **2 Private Subnets** in each VPC.
- Example (Hub):
 - **10.0.1.0/24** (AZ1)
 - **10.0.2.0/24** (AZ2)
 - Repeat for other VPCs.
3. **No Internet Gateway** (since we want private).
-

□ Step 2: Create Transit Gateways

1. In **each region**, go to **Transit Gateway Console** → **Create Transit Gateway**.
 - Name: **tgw-us-east-1** (Hub), **tgw-us-west-2**, etc.
 - Amazon side ASN: leave default (64512).
 - Multicast support: disable.
 - DNS support: enable.
 2. After creation, note down **TGW IDs** for all 4.
-

□ Step 3: Attach VPCs to TGWs

For each region:

1. Go to **Transit Gateway Attachments** → **Create Attachment**.

- Select your **TGW**.
- Attachment type: **VPC**.
- Select your VPC & private subnets.

2. Do this for all 4 VPCs.

□ **Step 4: Create TGW Peerings (Hub-and-Spoke)**

1. Go to **us-east-1 (Hub)** → **Transit Gateway Peering Attachments** → **Create**.

- Request Peering to **tgw-us-west-2**.
- Accept the request in **us-west-2**.

2. Repeat for **eu-west-1** and **ap-south-1**.

✓ Now TGWs are peered with Hub.

□ **Step 5: Update Route Tables**

VPC Route Tables:

- In each VPC's **Route Table**, add a route for other VPC CIDRs → point to TGW.

Example (Spoke1):

- Destination: **10.0.0.0/16** (Hub) → Target: **TGW**.
- Destination: **10.2.0.0/16** (Spoke2) → Target: **TGW**.
- Destination: **10.3.0.0/16** (Spoke3) → Target: **TGW**.

Do similarly for others.

TGW Route Tables:

- Hub TGW: Add routes for **all spoke CIDRs** pointing to their attachments.
 - Spokes TGW: Add route to **Hub CIDR** pointing to peering.
-

□ Step 6: Create VPC Endpoints

In each VPC, go to **VPC Console** → **Endpoints** → **Create Endpoint**:

1. **Gateway Endpoint:**

- Service: **S3**.
- Route table: Select your VPC route table.

2. **Interface Endpoints** (one by one):

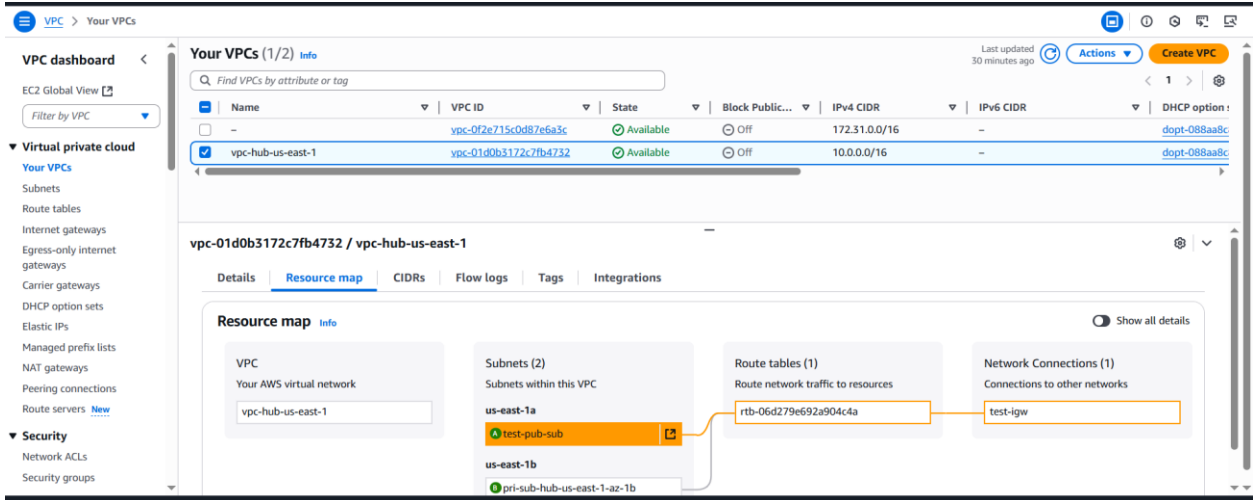
- **SSM**
 - **SSMMessages**
 - **EC2**
 - **ECR.api**
 - **ECR.dkr**
 - **CloudWatch Logs**
 - **Ensure private DNS is enabled.**
-

□ Step 7: Validation

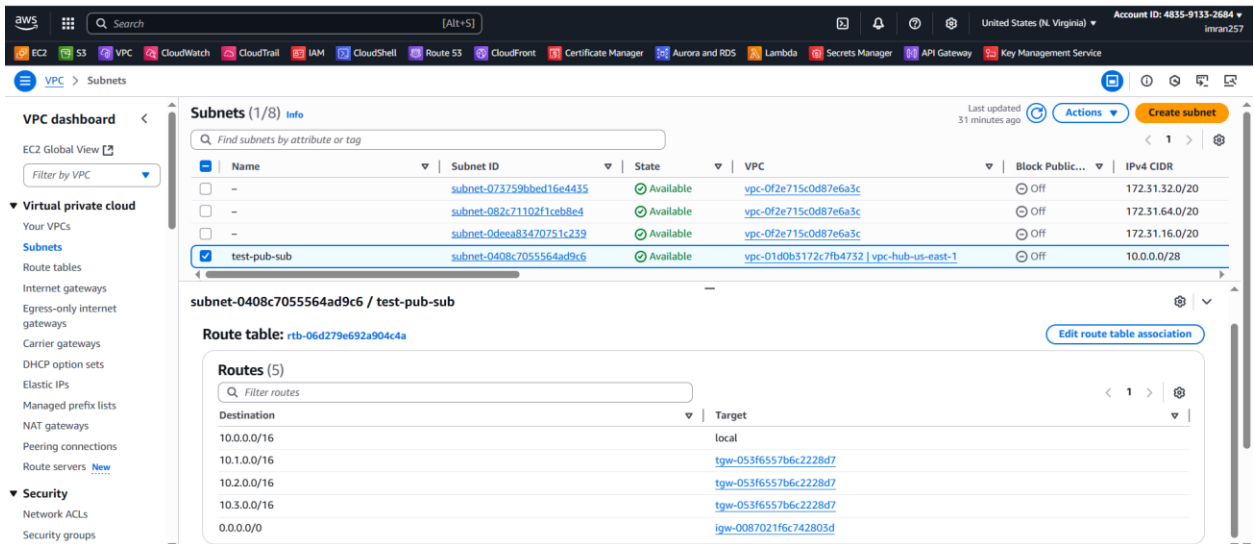
1. Launch **EC2 Instances** in private subnets (use SSM agent enabled AMI, like Amazon Linux 2/2023).
2. Connect via **SSM Session Manager** (no need for SSH key).
3. Validate connectivity:
 - Ping from Hub → Spoke VPCs.
 - Run `aws s3 ls` (should work via VPC endpoint).

- Check ECR & SSM access.

1. Created 4 VPC's in Virginia, Oregon, Ireland , Mumbai region having vpc range
2. 10.0.0.0 , 10.1.0.0, 10.2.0.0 10.3.0.0
3. Showing vpc of Virginia. Connect igw to treat as primary source to connect other regions ec2.



- 4.
5. Create subnets make routes to the all region destinations.



- 6.
7. Create route tables and make destinations

VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Route tables (1/2) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner
-	rtb-0c19279c219d8ec05	-	-	Yes	vpc-0f2e715cd87e6a3c	48:
-	rtb-06d279e692a904c4a	-	-	Yes	vpc-01d0b3172c7fb4732 vpc-hub-us-east-1	48:

rtb-06d279e692a904c4a

Details | **Routes** | Subnet associations | Edge associations | Route propagation | Tags

Routes (5)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	tgw-0087021f6c742803d	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
10.1.0.0/16	tgw-053f6557b6c2228d7	Active	No	Create Route
10.2.0.0/16	tgw-053f6557b6c2228d7	Active	No	Create Route
10.3.0.0/16	tgw-053f6557b6c2228d7	Active	No	Create Route

- 8.
9. Create transit gateway in all regions

VPC > Transit gateways

Transit gateways (1/1) Info

Find transit gateway by attribute or tag

Name	Transit gateway ID	State
tgw-us-east-1	tgw-053f6557b6c2228d7	Available

Transit gateway: tgw-053f6557b6c2228d7 / tgw-us-east-1

Details | Flow logs | Sharing | Tags

Details

Transit gateway ID tgw-053f6557b6c2228d7	Transit gateway ARN arn:aws:ec2:us-east-1:483591332684:transit-gateway/tgw-053f6557b6c2228d7	Owner ID 483591332684	Description verginia
State Available	Default association route table Enable	Default propagation route table Enable	Transit gateway CIDR blocks -
Amazon ASN	Association route table ID	Propagation route table ID	Security Group Referencing support

- 10.
11. Create attachments 1 is VPC connection for self and 3 Peering connections to other regions.need to send request and other regions need to accept.

VPC > Transit gateway attachments

Transit gateway attachments (7) Info

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
virg-mumbai	tgw-attach-01f646ac9d47e9abe	tgw-053f6557b6c2228d7	Available	Peering	tgw-0c56f70fe649efe5d
tg-attach-us-east-1-to-eu-west-1	tgw-attach-0379c6d7a8b3aecd	tgw-053f6557b6c2228d7	Deleted	Peering	tgw-072e35900916b813b
virg-ireland	tgw-attach-0a983dcaa38f34f1c	tgw-053f6557b6c2228d7	Available	Peering	tgw-072e35900916b813b
tg-attach-us-east-1-to-us-west-2	tgw-attach-0b4040d5fc7e7113d	tgw-053f6557b6c2228d7	Deleted	Peering	tgw-0c4d86d28c8d07ed5
tg-attach-us-east-1-to-ap-south-1	tgw-attach-0de3603d10ad8d37e	tgw-053f6557b6c2228d7	Deleted	Peering	tgw-0c56f70fe649efe5d
virg-to-oregon	tgw-attach-0fb759f29e6d4d6463	tgw-053f6557b6c2228d7	Available	Peering	tgw-0c4d86d28c8d07ed5
tg-attach-virginia	tgw-attach-0e31803ad4b7391dd	tgw-053f6557b6c2228d7	Available	VPC	vpc-01d0b3172c7fb4732

- 12.
13. Create transit gateway route tables with other regions use their CIDR.

VPC > Transit gateway route tables

Transit gateway route tables (1/1) info

Find transit gateway route table by attribute or tag

Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table
tgw-rtb-009a1367e7a0fe8bb	tgw-rtb-009a1367e7a0fe8bb	tgw-053f6557b6c2228d7	Available	Yes

Transit gateway route tables: tgw-rtb-009a1367e7a0fe8bb

Routes (4) info

Find route by attribute or tag

CIDR	Attachment ID	Resource ID	Resource type	Route type	Route state	P...
10.0.0.0/16	tgw-attach-0e31803ad4b7391dd	vpc-01d0b3172c7fb4732	VPC	Propagated	Active	-
10.1.0.0/16	tgw-attach-0fb759f29e64d6463	tgw-0c4d86d28c8d07ed5	Peering	Static	Active	-
10.2.0.0/16	tgw-attach-0a983dcaa38f34f1c	tgw-072e35900916b813b	Peering	Static	Active	-
10.3.0.0/16	tgw-attach-01f646ac9d47e9abe	tgw-0c56f70fe649efe5d	Peering	Static	Active	-

- 14.
15. Create EC2 and edit inbound rules to allow traffic rules in security group.

EC2 > Instances

Instances (1/1) info

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
useast1-ec2	i-0c5225c1dc8f4232b	Running	t2.nano	2/2 checks passed	View alarms +	us-east-1a	ec2-34-207-73-249.co...

i-0c5225c1dc8f4232b (useast1-ec2)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary info

Instance ID i-0c5225c1dc8f4232b	Public IPv4 address 34.207.73.249 open address	Private IPv4 addresses 10.0.0.9
IPv6 address -	Instance state Running	Public DNS ec2-34-207-73-249.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-0-0-9.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-0-9.ec2.internal	Elastic IP addresses
Answer private resource DNS name	Instance type	

16.

EC2 > Security Groups

sg-0550794937d9e8026 - default

Details

Security group name default	Security group ID sg-0550794937d9e8026	Description default VPC security group	VPC ID vpc-01d0b3172c7fb4732
Owner 483591332684	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing - new | VPC associations - new | Tags

Inbound rules (2)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-08b452ed4288e22b0	IPv4	SSH	TCP	22	0.0.0.0/0
-	sgr-0daa23edbb273c2aa	IPv4	All traffic	All	All	0.0.0.0/0

17.

Oregon

1. Created vpc in oregon region. And created end point only here to access the S3 Resource or S3 bucket of another region.

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Your VPCs (1/2)

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
-	vpc-0e7f5ff0401654d5c	Available	Off	172.31.0.0/16	-	dopt-03d3dc38b
✓ vpc-us-west-2	vpc-02b30fe1ca290a82d	Available	Off	10.1.0.0/16	-	dopt-03d3dc38b

vpc-02b30fe1ca290a82d / vpc-us-west-2

Resource map

VPC: Your AWS virtual network
vpc-us-west-2

Subnets (2): Subnets within this VPC
us-west-2a: pri-sub-us-west-2-az-2a
us-west-2b: pri-sub-us-west-2-az-2b

Route tables (1): Route network traffic to resources
rtb-01cfd58dd30d7d73

Network Connections (1): Connections to other networks
oregion-endpoint-gw

2. Created route table and attached routes of other regions.

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Route tables (1/2)

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Own.
✓	rtb-01cfd58dd30d7d73	-	-	Yes	vpc-02b30fe1ca290a82d vpc-us-west-2	4835
-	rtb-0c3af109db8caf24c	-	-	Yes	vpc-0e7f5ff0401654d5c	4835

rtb-01cfd58dd30d7d73

Routes (4)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	tgw-0c4d86d28c8d07ed5	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	tgw-0c4d86d28c8d07ed5	Active	No	Create Route
10.3.0.0/16	tgw-0c4d86d28c8d07ed5	Active	No	Create Route

3. Created transit gateway and attachments with resource type 1 VPC for self and 1 peering with virginia.

Transit gateways (1/1) info

Find transit gateway by attribute or tag

Name	Transit gateway ID	State
tgw-us-west-2	tgw-0c4d86d28c8d07ed5	Available

Transit gateway: tgw-0c4d86d28c8d07ed5 / tgw-us-west-2

Details | Flow logs | Sharing | Tags

Details

Transit gateway ID tgw-0c4d86d28c8d07ed5	Transit gateway ARN arn:aws:ec2:us-west-2:483591332684:transit-gateway/tgw-0c4d86d28c8d07ed5	Owner ID 483591332684	Description oregon
State Available	Default association route table Enable	Default propagation route table Enable	Transit gateway CIDR blocks -
Amazon ASN	Association route table ID	Propagation route table ID	Security Group Referencing support

Transit gateway attachments (5) info

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID	Ass
tg-attach-uswest2-to-eu-west-1	tgw-attach-036321b66d607a890	tgw-0c4d86d28c8d07ed5	Deleted	Peering	tgw-072e35900916b813b	-
tg-attach-us-west2-to-useast1	tgw-attach-0b4040d5fc7e7113d	tgw-0c4d86d28c8d07ed5	Deleted	Peering	tgw-053f6557b6c2228d7	-
tg-attach-uswest2-to-apsouth-1	tgw-attach-0c328c0416cf72ffb	tgw-0c4d86d28c8d07ed5	Deleted	Peering	tgw-0c56f70fe649efe5d	-
	tgw-attach-0fb759f29e64d6463	tgw-0c4d86d28c8d07ed5	Available	Peering	tgw-053f6557b6c2228d7	tgw
tg-attach-us-west-2	tgw-attach-05a152a4aae31eacc	tgw-0c4d86d28c8d07ed5	Available	VPC	vpc-02b30fe1ca290a82d	tgw

Select a transit gateway attachment

4. Adding associations with self VPC and peering to virginia. And i route tables add routes to all regions.

Transit gateway route tables (1/1) info

Find transit gateway route table by attribute or tag

Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagation...
tgw-rtb-03b8746d350571691	tgw-rtb-03b8746d350571691	tgw-0c4d86d28c8d07ed5	Available	Yes	Yes

Transit gateway route tables: tgw-rtb-03b8746d350571691

Details | Associations | Propagations | Prefix list references | Routes | Tags

Associations (2) info

Find association by attribute or tag

Attachment ID	Resource type	Resource ID	State
tgw-attach-0fb759f29e64d6463	Peering	tgw-053f6557b6c2228d7	Associated
tgw-attach-05a152a4aae31eacc	VPC	vpc-02b30fe1ca290a82d	Associated

Transit gateway route tables (1/1)

Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagati...
☒	tgw-rtb-03b8746d350571691	tgw-0c4d86d28c8d07ed5	Available	Yes	Yes

Transit gateway route tables: tgw-rtb-03b8746d350571691

Routes (4)

☐	CIDR	Attachment ID	Resource ID	Resource type	Route type	Route state	P...
☐	10.0.0/16	tgw-attach-0fb759f29e64d6463	tgw-053f6557b6c2228d7	Peering	Static	Active	-
☐	10.1.0/16	tgw-attach-05a152a4ae31eacc	vpc-02b30fe1ca290a82d	VPC	Propagated	Active	-
☐	10.2.0/16	tgw-attach-0fb759f29e64d6463	tgw-053f6557b6c2228d7	Peering	Static	Active	-
☐	10.3.0/16	tgw-attach-0fb759f29e64d6463	tgw-053f6557b6c2228d7	Peering	Static	Active	-

In same way the above things need to do in other regions also,

1. Now created EC2 with having igw as a primary source. And in inbound rules of security group allow ssh and all traffic.

The screenshot shows the AWS Management Console interface. The left sidebar contains navigation links for Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, and Network & Security. The main content area displays the 'Instances (1/1)' page. A table lists the instance 'uswest2-ec2' with ID 'i-0dab1346536bfc99', state 'Running', and type 't2.nano'. Below the table, the 'Details' tab is active, showing instance summary information including the instance ID, IP addresses, hostname type, and public/private IP addresses. A tooltip indicates that the private IPv4 address has been copied.

The screenshot shows the AWS Management Console interface. The left sidebar contains navigation links for Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, and Network & Security. The main content area displays the 'Security Groups' page. A green notification banner at the top states 'Inbound security group rules successfully modified on security group (sg-0ed208faa82e2badd | default)'. Below this, the 'Details' tab is active for the security group 'sg-0ed208faa82e2badd - default'. The details section shows the security group name, ID, description, VPC ID, owner, and rule counts. Below the details, the 'Inbound rules (2)' section is visible, showing a table with two rules: one for SSH access on port 22 and another for all traffic.

2. Created instances in other regions as private having no internet.

TEST:

1. Now tried to ping from VIRGINIA to OREGON


```

[ec2-user@ip-10-0-0-9 ~]$ ping 10.3.2.126
PING 10.3.2.126 (10.3.2.126) 56(84) bytes of data.
64 bytes from 10.3.2.126: icmp_seq=59 ttl=124 time=204 ms
64 bytes from 10.3.2.126: icmp_seq=60 ttl=124 time=203 ms
64 bytes from 10.3.2.126: icmp_seq=61 ttl=124 time=203 ms
64 bytes from 10.3.2.126: icmp_seq=62 ttl=124 time=203 ms
64 bytes from 10.3.2.126: icmp_seq=63 ttl=124 time=203 ms
64 bytes from 10.3.2.126: icmp_seq=64 ttl=124 time=203 ms
64 bytes from 10.3.2.126: icmp_seq=65 ttl=124 time=203 ms
^C
--- 10.3.2.126 ping statistics ---
65 packets transmitted, 7 received, 89.2308% packet loss, time 66345ms
rtt min/avg/max/mdev = 202.832/203.103/204.258/0.473 ms
[ec2-user@ip-10-0-0-9 ~]$

```

ACCESSING S3 RESOURCE using ENDPOINT without Internet.

1. End point created in oregon

The screenshot shows the AWS Management Console for the 'oregon' region. The 'Endpoints' page is displayed, showing a table of endpoints. The 'oregeion-endpoint-gw' endpoint is selected, and its details are shown below.

Name	VPC endpoint ID	Endpoint type	Status	Service name
ep-us-west2	vpce-07ae4fc91e5f4f393	Interface	Available	com.amazonaws.us-west-2.s3
oregeion-endpoint-gw	vpce-0211e9be57b5e2303	Gateway	Available	com.amazonaws.us-west-2.s3

vpce-0211e9be57b5e2303 / oregeion-endpoint-gw

Details	Status	Creation time	Endpoint type
Endpoint ID vpce-0211e9be57b5e2303	Available	Wednesday, August 20, 2025 at 19:43:28 GMT+5:30	Gateway
VPC ID vpce-02b30fe1ca290a82d (vpc-us-west-2)	Status message -	Service name com.amazonaws.us-west-2.s3	Private DNS names enabled No
Service region us-west-2			

VPC dashboard < Your VPCs (1/2) info

EC2 Global View [\[?\]](#)

Filter by VPC [▼](#)

▼ **Virtual private cloud**

Your VPCs

- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

▼ **Security**

- Network ACLs
- Security groups

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
-	vpc-De7f5ff040165dd5c	Available	Off	172.31.0.0/16	-	dopt-03d3dc38b
☒ vpc-us-west-2	vpc-02b30fe1ca290a82d	Available	Off	10.1.0.0/16	-	dopt-03d3dc38b

▼ vpc-02b30fe1ca290a82d / vpc-us-west-2

VPC
Your AWS virtual network
vpc-us-west-2

Subnets (2)
Subnets within this VPC
us-west-2a
pri-sub-us-west-2-az-2a
us-west-2b
pri-sub-us-west-2-az-2b

Route tables (1)
Route network traffic to resources
rtb-01cfd58dd30d7d73

Network Connections (1)
Connections to other networks
oregon-endpoint-gw

2. Created s3 in mumbai region

Amazon S3

General purpose buckets | All AWS Regions | Directory buckets

General purpose buckets (1/1) info

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
imran-mumbai-s3	Asia Pacific (Mumbai) ap-south-1	August 20, 2025, 19:48:45 (UTC+05:30)

Account snapshot info [View dashboard](#)

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary - new info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

3. Login ec2 of virginia and make this as jump server , taken pem key of ec2 of oregon and then login to ec2 of oregon which is have private ip and now configure aws and

Aws s3 ls

```
[ec2-user@ip-10-0-0-9 ~]$ vi oregon.pem
[ec2-user@ip-10-0-0-9 ~]$ chmod 600 oregon.pem
[ec2-user@ip-10-0-0-9 ~]$ ssh -i oregon.pem ec2-user@10.1.1.46
The authenticity of host '10.1.1.46 (10.1.1.46)' can't be established.
ED25519 key fingerprint is SHA256:kwOZawj0xNpYF3lwpNKjFCU/pn3ueZlkydy1tfsgpFY.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.1.1.46' (ED25519) to the list of known hosts.
```

```
      #_
    _/  #####_      Amazon Linux 2023
   ~~~ \#####\
   ~~~  \####|
   ~~~   \###|
   ~~~    \#/  _/
   ~~~     V~' '->
   ~~~
   ~~~  _/
   ~~~ _/
   ~~~ _/m/ '->
```

```
[ec2-user@ip-10-1-1-46 ~]$ asw config
-bash: asw: command not found
[ec2-user@ip-10-1-1-46 ~]$ aws configure
AWS Access Key ID [None]: AKIAXBGCKENGJYGN6NF6
AWS Secret Access Key [None]: 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
Default region name [None]:
Default output format [None]:
[ec2-user@ip-10-1-1-46 ~]$ aws s3 ls
2025-08-20 14:18:47 imran-mumbai-s3
[ec2-user@ip-10-1-1-46 ~]$
```